

Morphic Studio Software Architecture Specification

Document 50 — Observability, Telemetry & Operational Intelligence Architecture

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1. Purpose

This document defines the Observability, Telemetry, Monitoring, and Operational Intelligence Architecture for Morphic Studio.

Its purpose is to provide complete visibility into platform behavior, application health, AI operations, workflows, infrastructure, and user experience while enabling proactive operational decision-making.

2. Observability Philosophy

The platform follows six principles.

2.1 Observe Everything

Every significant action within the platform should produce observable signals.

These include:

- User interactions
 - API requests
 - AI operations
 - Workflow execution
 - Infrastructure events
 - Plugin activity
 - Background jobs
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2.2 Correlation Over Isolation

Individual events are useful.

Correlated events provide operational intelligence.

Every subsystem should contribute to a unified operational view.

2.3 Privacy by Design

Observability must respect user privacy.

Sensitive information should never appear in telemetry.

Personally identifiable information should be minimized, protected, or anonymized where appropriate.

2.4 Actionable Metrics

Metrics should support operational decisions rather than merely collecting data.

2.5 Continuous Learning

Operational insights should improve future deployments, workflows, AI routing, and platform performance.

2.6 AI-Assisted Operations

AI may analyze operational data to identify anomalies, trends, optimization opportunities, and probable root causes.

Human operators remain responsible for operational decisions.

3. Telemetry Sources

Telemetry is collected from:

Frontend

Backend

Database

AI Gateway

Workflow Engine

Plugin System

Infrastructure

Background Workers

Search Services

Object Storage

Rendering Services

Authentication

Marketplace

4. Structured Logging

Every log entry includes:

Timestamp

Severity

Service

Environment

Correlation ID

Request ID

Workflow ID (if applicable)

AI Request ID (if applicable)

Plugin ID (if applicable)

User ID (when appropriate)

Logs should be structured, searchable, and retained according to operational policies.

5. Metrics Architecture

Collect metrics for:

API performance

Database queries

Workflow execution

AI latency

Token consumption

Rendering duration

Queue depth

CPU

Memory

Storage

Network

Plugin performance

Marketplace usage

Metrics should support dashboards and historical analysis.

6. Distributed Tracing

Trace requests across:

Frontend

API

Authentication

Workflow Engine

AI Gateway

Database

Plugins

Background Workers

External Services

Every trace receives a globally unique Trace ID.

7. Health Monitoring

Monitor:

Application availability

API responsiveness

Database health

Queue health

AI provider availability

Storage systems

Search services

Rendering services

Authentication systems

Background jobs

Health checks should support automated recovery where appropriate.

8. Operational Intelligence Layer

The Operational Intelligence Layer correlates:

Logs

Metrics

Traces

Workflow events

AI events

Plugin events

Infrastructure events

User interaction patterns

The layer produces actionable operational insights rather than isolated data points.

9. AI Performance Analytics

Measure:

Model latency

Response quality

Fallback frequency

Capability usage

Provider reliability

Token efficiency

Cost per capability

Context retrieval effectiveness

These analytics guide AI optimization and provider selection.

10. Workflow Analytics

Track:

Execution duration

Success rate

Retry frequency

Approval delays

Automation adoption

Resource utilization

Workflow complexity

Failure patterns

Insights support workflow optimization.

11. User Experience Analytics

Measure:

Page responsiveness

Navigation performance

Feature adoption

Session duration

Task completion

Interaction latency

Accessibility metrics

User feedback

Analytics must remain privacy-aware.

12. Capacity Planning

Forecast future resource requirements using:

Growth trends

Traffic patterns

Storage usage

Database growth

AI consumption

Workflow volume

Rendering demand

Capacity planning should support proactive scaling.

13. Cost Intelligence

Track operational costs including:

Infrastructure

Storage

Bandwidth

AI providers

Rendering

Search

Monitoring

Third-party integrations

Provide dashboards and forecasting tools.

14. Alerting & Incident Intelligence

Alerts should include:

Severity

Affected services

Correlated events

Probable root cause

Recommended actions

Escalation path

Historical context

Alert fatigue should be minimized through intelligent correlation.

15. Reporting & Dashboards

Provide dashboards for:

System Health

Application Performance

AI Operations

Workflow Automation

Infrastructure

Security

Marketplace

Business Metrics

Executive Summary

Dashboards should support different operational roles.

16. Acceptance Criteria

The observability architecture is successful if:

- Every critical subsystem is observable.
 - Logs, metrics, and traces are correlated.
 - Operational insights support decision-making.
 - AI operations remain measurable.
 - Privacy is preserved.
 - Capacity planning is data-driven.
 - Operational intelligence enables proactive management.
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Conclusion

The Morpheo Studio Observability, Telemetry, and Operational Intelligence Architecture establishes a comprehensive operational awareness framework that extends beyond traditional monitoring. By combining structured telemetry, distributed tracing, AI analytics, workflow intelligence, and predictive operational insights, the platform gains the visibility required to operate reliably, optimize continuously, and scale confidently as a modern AI-native creative ecosystem.